

Infrastructure Recommendation Executive Brief

Ardent ISD | Ardent High School Wireless Modernization

Decision Requested

Authorize the technology department to proceed with a phased replacement and redesign of Ardent High School wireless infrastructure, with final procurement contingent on technical validation, current E-Rate eligibility review, and district purchasing requirements.

Why This Is Necessary

Ardent High School has the district's highest concentration of wireless performance complaints. The issue is not simply internet bandwidth. Peak internet usage remains below the district's available capacity, while classroom-level reports and monitoring indicate inconsistent performance in high-density areas.

Evidence

Evidence	Finding
Help desk	Wireless/connectivity tickets at AHS are 2.3x the district secondary-campus average.
Access point lifecycle	68 of 146 AHS access points are in or approaching the district replacement window.
High-density areas	Library, cafeteria, fine arts wing, and two science corridors show the most consistent congestion reports.
Internet utilization	District peak internet use is approximately 61% of primary capacity, suggesting the issue is not solely the ISP circuit.
Growth	AHS enrollment is projected to increase by approximately 6% over three years.

Options Considered

Option	Advantages	Limitations	Planning Estimate
A. Replace only failed/oldest access points	Lowest immediate cost	Preserves inconsistent design; likely extends support burden	\$78,000
B. Phased redesign and refresh	Targets density, lifecycle, cabling, and monitoring as one plan	Requires planned implementation window and more coordination	\$184,000
C. Full one-time replacement	Fastest standardization	Largest single-year budget impact; may replace some serviceable equipment early	\$248,000

Recommendation

Proceed with **Option B: phased redesign and refresh**.

This option best balances instructional reliability, lifecycle management, staff capacity, and budget impact. The project should include validation of access point placement, switching capacity, cabling, power, MDF/IDF readiness, monitoring, and implementation sequencing.

Risks and Dependencies

- E-Rate eligibility must be verified against current program rules.
- Work windows must avoid state testing and major instructional periods.
- Facilities coordination is required for cabling, electrical, ceiling access, and environmental conditions.
- The district should confirm expected support and licensing costs for the full lifecycle, not only acquisition cost.

Success Measures

- 50% reduction in wireless-related tickets at AHS within one semester of completion.
- No recurring high-density coverage gaps identified in post-installation validation.
- All installed equipment entered into asset inventory with location, warranty, and replacement year.
- Network monitoring baseline established for utilization and availability.

Reference Points

- [CoSN IT Management Resources](#) - K-12 technology operations and leadership resources.
- [USAC E-Rate Program](#) - Federal E-Rate program information for schools and libraries.
- [SETDA Infrastructure for the Future](#) - Guidance for future-ready broadband and infrastructure planning.
- [Future Ready Schools Frameworks](#) - Systems-oriented planning connecting infrastructure, learning, leadership, budget, and equity.

Leadership use: This document is designed to support district-level review, discussion, and decision-making. Technical validation, procurement review, legal review, or compliance review should be completed by the appropriate district specialists when applicable.